Chapter 1

Ordinary Differential Equations

Consider the ordinary differential equation

$$y'(t) = f(t, y, u(t, y)), \tag{1.1}$$

where u is a smooth system input or characteristic map. Given an approximation $\tilde{u}$ of u, we are interested in finding estimates or bounds for the local and global error of a numerical integration scheme, e.g.

$$\|\tilde{y}(T) - y_N\|, \quad \|\tilde{y}(h) - y_1\|, \tag{1.2}$$

where $T > 0, y_n$ denote the numerical solutions after n steps and $\tilde{y}$ denotes the solution of the perturbed equation

$$y' = f(t, y, \tilde{u}(t, y)). \tag{1.3}$$

Specifically, we only consider problems where f is Lipschitz in both y and u and $\tilde{u}$ is Lipschitz continuous in y, making both equations uniquely solvable for a given initial value $y(0) = y_0$. Classical error analysis may be insufficient here, since $\tilde{u}$ does not have to satisfy the typical smoothness assumptions for the right-hand side regarding differentiability or boundedness of higher order derivatives.

1.1 One-Step methods

Since we specifically want to consider system inputs which dependent both on time and state, we require the following simple extension of Taylor's theorem.

Lemma 1. Let $f : X \to F, X \subset \mathbb{R}^m, F \subset \mathbb{R}^n$, be in the class $C^q(X, F)$, such that $\partial^q f(x)$ is Lipschitz continuous in x. Then if $h \in \mathbb{R}^m$ and $x + th \in X$ for all $t \in [0, 1]$, we have

$$f(x + h) = \sum_{k=0}^{q} \frac{1}{k!} \partial^k f(x)[h]^k + R(x, h), \quad \|R(x, h)\| \leq C\|h\|^{q+1}.$$

Proof. For $q \geq 1$, this follows directly from [AmannEscher, Analysis II, Remark 5.9]. For $q = 0$ ... □

We will now derive the local error for a one-step method applied to the system in (1.3).

Theorem 1. If

- f is sufficiently smooth, Lipschitz continuous in y and u and bounded
- $\tilde{u} \in C^q$, $\tilde{u}$ and $\partial^q \tilde{u}$ Lipschitz continuous in (t, y), $\tilde{u}$ bounded,

then a one-step method of order p

$$y_{n+1} = y_n + h_n \Phi(t_n, y_n, h_n, f, u),$$

where Φ is Lipschitz continuous in u, yields the local error

$$\tilde{y}(t_{n+1}) - (\tilde{y}(t_n) + h_n \Phi(t_n, \tilde{y}(t_n), h_n, f, \tilde{u})) = \mathcal{O}(h^{q+2}) + \mathcal{O}(h^{p+1}), \quad 0 < h_n \leq h$$

Proof. Following [Arnold 2010], write the local error as

$$\begin{aligned} le_n = \tilde{y}(t_{n+1}) &- \hat{y}(t_{n+1}) + \hat{y}(t_{n+1}) \\ &- \hat{y}_{n+1} + \hat{y}_{n+1} \\ &- (\tilde{y}(t_n) + h_n \Phi(t_n, \tilde{y}(t_n), h_n, f, \tilde{u})), \end{aligned}$$

where $\hat{y} = \hat{y}(t)$ and

$$\hat{y}_{n+1} = \tilde{y}(t_n) + h_n \Phi(t_n, \tilde{y}(t_n), h_n, f, \hat{u}_n)$$

are the exact and one-step numerical solution of the locally defined smooth shadow problem

$$y'(t) = f(t, y(t), \hat{u}_n(t, y(t))), \quad y(t_n) = \tilde{y}(t_n) \tag{1.4}$$

for smooth system input $\hat{u}_n$ obtained from the truncated multivariate Taylor polynomial, i.e.

$$\hat{u}_n(t, y) = \sum_{k=0}^{q} \frac{1}{k!} \partial^k \tilde{u}(t_n, \tilde{y}(t_n)) \begin{bmatrix} t - t_n \\ y - \tilde{y}(t_n) \end{bmatrix}^k. \tag{1.5}$$

Since f is Lipschitz in u, we get

$$\|f(t, \hat{y}(t), \tilde{u}(t, \hat{y}(t))) - f(t, \hat{y}(t), \hat{u}_n(t, \hat{y}(t)))\| \leq L_{f,u} \|\tilde{u}(t, \hat{y}(t)) - \hat{u}_n(t, \hat{y}(t))\|.$$

Therefore, by means of classical error analysis [HNW] and Taylor's theorem (1),

$$\begin{aligned} \|\tilde{y}(t_{n+1}) - \hat{y}(t_{n+1})\| &\leq C_1 h \sup_{t \in [t_n, t_{n+1}]} \left\| \begin{pmatrix} t - t_n \\ \hat{y}(t) - y(t_n) \end{pmatrix} \right\|^{q+1} \\ &\leq C_2 h^{q+2}, \end{aligned} \tag{1.6}$$

where the last line follows from the fact that $\hat{y}(t)$ is the solution of (1.4) and f is bounded (since $\hat{u}_n$ as the truncated Taylor polynomial of $\tilde{u}$ is bounded). Additionally, for a constant C_3 independent of n,

$$\|\hat{y}(t_{n+1}) - \hat{y}_{n+1}\| \leq C_3 h^{p+1}. \tag{1.7}$$

With a Lipschitz condition on Φ in u, we lastly obtain

$$\begin{aligned} \|\hat{y}_{n+1} - (\tilde{y}(t_n) + h_n \Phi(t_n, \tilde{y}(t_n), h_n, f, \tilde{u}))\| &\leq h L_{\Phi,u} \sup_{(t,y) \in S} \|\hat{u}_n(t, y) - \tilde{u}(t, y)\| \\ &\leq h C_4 \sup_{(t,y) \in S} \left\| \begin{pmatrix} t - t_n \\ y - y(t_n) \end{pmatrix} \right\|^{q+1} \\ &\leq C_5 h^{q+2} \end{aligned} \tag{1.8}$$

where S contains the stage vectors of the numerical integrator applied to (1.4). With (1.6), (1.7), (1.8) and the triangle inequality, we finally get

$$\|le_n\| = \mathcal{O}(h^{q+2}) + \mathcal{O}(h^{p+1}). \tag{1.9}$$

□

Example 1. For an s-stage Runge-Kutta method, we have

$$\Phi(t, y, h, f, u) = \sum_{i=1}^{s} b_i f(t + c_i h, Y_i, u(t + c_i h, Y_i)), \quad Y_i = y + h \sum_{j=1}^{s} a_{ij} f(t + c_j h, Y_j, u(t + c_j h, Y_j)). \tag{1.10}$$

Therefore

$$\Phi(t, y, h, f, \hat{u}) - \Phi(t, y, h, f, \tilde{u}) = \sum_{i=1}^{s} b_i \left(f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i)) \right) \tag{1.11}$$

omitting times $t + c_i h$ for better readability. Now

$$\begin{aligned} f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i)) = f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) &- f(\hat{Y}_i, \tilde{u}(\hat{Y}_i)) + f(\hat{Y}_i, \tilde{u}(\hat{Y}_i)) \\ &- f(\hat{Y}_i, \tilde{u}(\tilde{Y}_i)) + f(\hat{Y}_i, \tilde{u}(\tilde{Y}_i)) \\ &- f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i)), \end{aligned} \tag{1.12}$$

such that for $C := L_{f,u}L_{\tilde{u},y} + L_{f,y}$

$$\|f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i))\| \leq L_{f,u}\|\hat{u}(\hat{Y}_i) - \tilde{u}(\hat{Y}_i)\| + C\|\hat{Y}_i - \tilde{Y}_i\|.$$

Notice that

$$\|\hat{Y}_i - \tilde{Y}_i\| \leq h\sum_{j=1}^{s} |a_{ij}|\|f(\hat{Y}_j, \hat{u}(\hat{Y}_j)) - f(\tilde{Y}_j, \tilde{u}(\tilde{Y}_j))\|$$

and by taking the maximum with respect to i, we get for $C_a := C \cdot \max_i \sum_{j=1}^{s} |a_{ij}|$

$$\max_i \|f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i))\| \leq L_{f,u} \max_i \|\hat{u}(\hat{Y}_i) - \tilde{u}(\hat{Y}_i)\| + hC_a \max_i \|f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i))\|,$$

showing that

$$\begin{aligned}\|\Phi(t, y, h, f, \hat{u}) - \Phi(t, y, h, f, \tilde{u})\| &\leq \left(\sum_{i=1}^{s} |b_i|\right) \max_i \|f(\hat{Y}_i, \hat{u}(\hat{Y}_i)) - f(\tilde{Y}_i, \tilde{u}(\tilde{Y}_i))\| \\ &\leq \frac{L_{f,u} \max_i \|\hat{u}(\hat{Y}_i) - \tilde{u}(\hat{Y}_i)\|}{1 - hC_a}.\end{aligned}$$

We obtain the requirement $h \leq 1/C_a$.